

Chemistry Short Answer Questions

UNIT-1

1. Define Heisenberg uncertainty principle.
2. Write the bond order of CO molecule.
3. Write the significance of Ψ and Ψ^2 .
4. Calculate the bond order in CO molecule.
5. Differences between bonding and anti-bonding molecular orbitals.
6. Draw the π -molecular orbitals of Benzene.
7. Why is oxygen paramagnetic? Give reasons.
8. Draw the HOMO and LUMO orbitals of Benzene.
9. Draw the HOMO and LUMO orbitals of 1,3-butadiene.
10. Derive de Broglie wave equation.

UNIT-2

1. What are the types of elemental semiconductors?
2. How are HPC nanomaterials utilized?
3. Write the applications of semiconductors in various fields.
4. Discuss the applications of HPC and core-shell nanomaterials.
5. Write the applications of fullerene.
6. How is a supercapacitor advantageous over a battery?
7. What is meant by critical temperature?
8. What is doping? List the dopants used.
9. How are nanomaterials classified?

UNIT-3

1. Write any four applications of graphene.
2. Distinguish between a primary cell and a secondary cell.
3. List the applications of PEMFCs.
4. What are the different kinds of electrochemical sensors?
5. What is an electrochemical cell?
6. List the advantages of potentiometric titrations.
7. Write equations of zinc-air battery.
8. What is a potentiometric sensor? Give an example.
9. Identify the technology used in PEMFCs. Give an example of a polymer electrolyte.
10. List the advantages of conductometric titrations.
11. Write the Nernst equation and explain its terms.
12. Define a fuel cell.

13. Write the applications of Lithium-ion battery.

UNIT-4

1. What is meant by functionality of monomer?
2. What are the catalysts used in coordination polymerization?
3. List any four differences between thermoplastics and thermosetting plastics.
4. Identify the monomers of: (i) Nylon-6,6 (ii) Polyacrylonitrile.
5. Identify the monomers of: (a) Buna-S (b) Polyglycolic acid.
6. List the applications of conducting polymers.
7. List any four differences between addition and condensation polymerization.
8. Mention the applications of PVC in various fields.

UNIT-5

1. Write the principle of UV-Visible spectroscopy.
2. Write the selection rules in IR spectroscopy.
3. What are the different types of electromagnetic radiation?
4. How many vibrational modes are there in linear and non-linear molecules having n atoms?
5. What is chromatography? Write its principle.
6. State Beer-Lambert's law.
7. Justify: 'IR spectroscopy is called vibrational spectroscopy.'
8. Write any two electronic transitions in UV spectroscopy.
9. Write applications of IR spectroscopy.
10. List the stretching vibrations in IR spectroscopy.